# Comparing DDPG and TD3 on LunarLander

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## 1 Introduction

We use the continuous-action version of LunarLander-v3. The observation includes the lander's position, velocity, angle, angular velocity, and two indicators showing whether each leg is touching the ground. The action consists of two continuous components controlling the main and side engines. Rewards encourage a safe landing near the landing pad and penalize excessive speed, tilt, and engine use.

## 2 DDPG with and without layer normalization

We compare DDPG with and without layer normalization (LN) on LunarLander. Both the actor and critic networks used two hidden layers of 64 units with ReLU activations. In the LN configuration, layer normalization was applied after each hidden linear layer and before its activation in both networks.

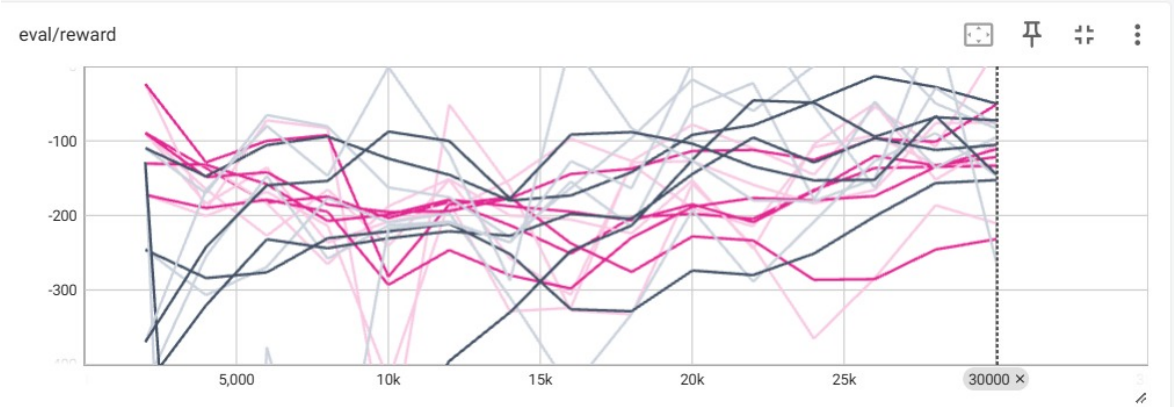
The target networks were initialized as copies of the actor and critic and updated through soft updates after each training update. The critic minimized the mean squared Bellman error, with bootstrapping disabled at terminal states. The hyperparameters were identical for both configurations and are listed in Table 1.

Table 1: DDPG hyperparameters.

| Setting | Value |
|---|---|
| Training steps per run and seeds per configuration | 30,000 and 5 |
| Number of training environments | 1 |
| Steps per update and steps before learning starts | 1 and 1,000 |
| Replay buffer size and batch size | 200,000 and 64 |
| Discount factor ($\gamma$) | 0.98 |
| Target network update coefficient ($\tau$) | 0.05 |
| Exploration noise standard deviation | 0.1 |
| Actor and critic learning rates | 0.001 for both |
| Evaluation interval (steps) and evaluation episodes | 2,000 and 10 |

We compare the results with and without layer normalization (LN). Table 2 summarizes all six recorded measures at the final checkpoint, using the raw values from the five seeds.

**Learning performance.** The reward curves fluctuate considerably and differ across seeds (Figure 1). The final mean reward was $-99.7$ without LN and $-133.1$ with LN. Final rewards ranged from $-210.4$ to 23.4 without LN, compared with $-262.7$ to $-78.4$ with LN. Although the mean was higher without LN, the large variation between runs makes it difficult to identify a consistent performance advantage. LN did not improve the average final reward in these runs.




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| ddpg-LunarLander-v3-LN-S0-20261012-204834 | -152.3786 | -146.2853 | 30,000 | 3.164 min |
| ddpg-LunarLander-v3-LN-S1-20261012-205151 | -49.9841 | -83.2578 | 30,000 | 3.048 min |
| ddpg-LunarLander-v3-LN-S2-20261012-205500 | -105.1933 | -94.8446 | 30,000 | 3.207 min |
| ddpg-LunarLander-v3-LN-S3-20261012-205820 | -72.2949 | -78.4444 | 30,000 | 3.31 min |
| ddpg-LunarLander-v3-LN-S4-20261012-210146 | -144.8573 | -262.6889 | 30,000 | 3.189 min |
| ddpg-LunarLander-v3-noLN-S0-20261012-203503 | -133.4804 | -131.8675 | 30,000 | 2.681 min |
| ddpg-LunarLander-v3-noLN-S1-20261012-203750 | -110.9365 | -75.283 | 30,000 | 2.644 min |
| ddpg-LunarLander-v3-noLN-S2-20261012-204035 | -231.4074 | -210.3922 | 30,000 | 2.472 min |
| ddpg-LunarLander-v3-noLN-S3-20261012-204309 | -121.3626 | -104.3547 | 30,000 | 2.527 min |
| ddpg-LunarLander-v3-noLN-S4-20261012-204549 | -51.2845 | 23.4438 | 30,000 | 2.641 min |

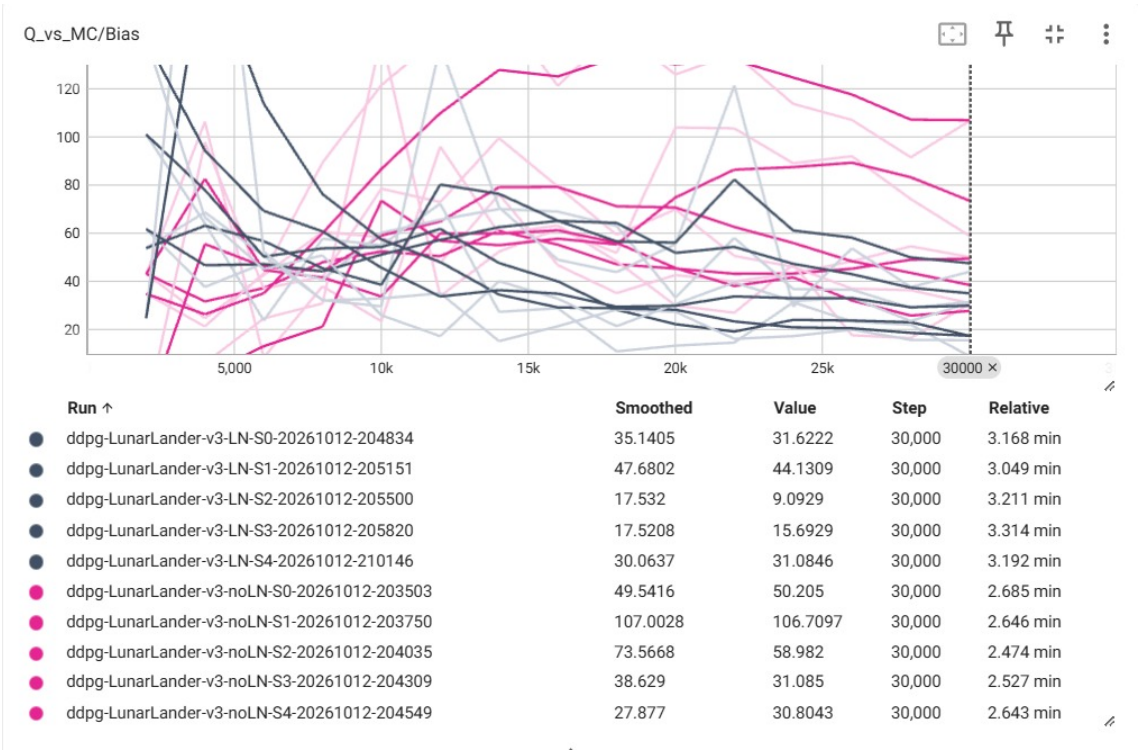
Figure 1: Evaluation reward versus environment steps for DDPG. Run labels identify the condition (LN/noLN) and seed (S0–S4). Bold and faint traces show TensorBoard-smoothed and raw values, respectively; they are not confidence bands. The supplied plot clips some extreme values.

Table 2: Final checkpoint at 30,000 steps for DDPG: mean ± sample standard deviation across five seeds. These are unsmoothed values; the loss entries describe the final training update.

| Measure | Without LN | With LN |
|---|---|---|
| Evaluation reward | $-99.7 \pm 85.2$ | $-133.1 \pm 77.3$ |
| Predicted $Q(s_0, \pi(s_0))$ | $74.9 \pm 26.5$ | $43.6 \pm 15.1$ |
| Monte Carlo return $G_0$ | $19.3 \pm 5.6$ | $17.3 \pm 4.6$ |
| Estimated bias $Q - G_0$ | $55.6 \pm 31.1$ | $26.3 \pm 13.9$ |
| Actor loss | $-70.1 \pm 23.9$ | $-39.4 \pm 16.4$ |
| Critic loss | $24.1 \pm 16.7$ | $4.0 \pm 1.7$ |

**Q-values and Monte Carlo returns.** Predicted Q-values rose early in training and finished higher without LN for every matched seed. Their final mean was 74.9 without LN and 43.6 with LN. Monte Carlo returns generally increased during training and were positive at the final checkpoint in all ten runs. Their final means were much closer: 19.3 without LN and 17.3 with LN. The larger Q-values without LN therefore did not correspond to similarly larger observed discounted returns.

**Overestimation bias.** The final difference $Q - G_0$ was positive in every run (Figure 2). Its mean fell from 55.6 without LN to 26.3 with LN, a reduction of about half. LN gave a smaller final bias in four of the five matched seed comparisons. The standard deviation was also lower with LN (13.9 versus 31.1), although the curves still show fluctuations during training.




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| ddpg-LunarLander-v3-LN-S0-20261012-204834 | 35.1405 | 31.6222 | 30,000 | 3.168 min |
| ddpg-LunarLander-v3-LN-S1-20261012-205151 | 47.6802 | 44.1309 | 30,000 | 3.049 min |
| ddpg-LunarLander-v3-LN-S2-20261012-205500 | 17.532 | 9.0929 | 30,000 | 3.211 min |
| ddpg-LunarLander-v3-LN-S3-20261012-205820 | 17.5208 | 15.6929 | 30,000 | 3.314 min |
| ddpg-LunarLander-v3-LN-S4-20261012-210146 | 30.0637 | 31.0846 | 30,000 | 3.192 min |
| ddpg-LunarLander-v3-noLN-S0-20261012-203503 | 49.5416 | 50.205 | 30,000 | 2.685 min |
| ddpg-LunarLander-v3-noLN-S1-20261012-203750 | 107.0028 | 106.7097 | 30,000 | 2.646 min |
| ddpg-LunarLander-v3-noLN-S2-20261012-204035 | 73.5668 | 58.982 | 30,000 | 2.474 min |
| ddpg-LunarLander-v3-noLN-S3-20261012-204309 | 38.629 | 31.085 | 30,000 | 2.527 min |
| ddpg-LunarLander-v3-noLN-S4-20261012-204549 | 27.877 | 30.8043 | 30,000 | 2.643 min |

Figure 2: Estimated bias $Q - G_0$ versus environment steps for the ten runs of DDPG. Each raw observation uses one Monte Carlo episode. Some peaks exceed the supplied plot's vertical range.

**Actor and critic losses.** The actor loss became more negative early in training and then continued to fluctuate. Its final mean was $-70.1$ without LN and $-39.4$ with LN. Because this loss is the negative mean predicted Q-value on a training batch, a more negative loss does not by itself indicate a better policy. The critic loss showed frequent spikes. At the final update, it was lower with LN for all five matched seeds, averaging 4.0 compared with 24.1 without LN. These endpoint values do not establish stability throughout training.

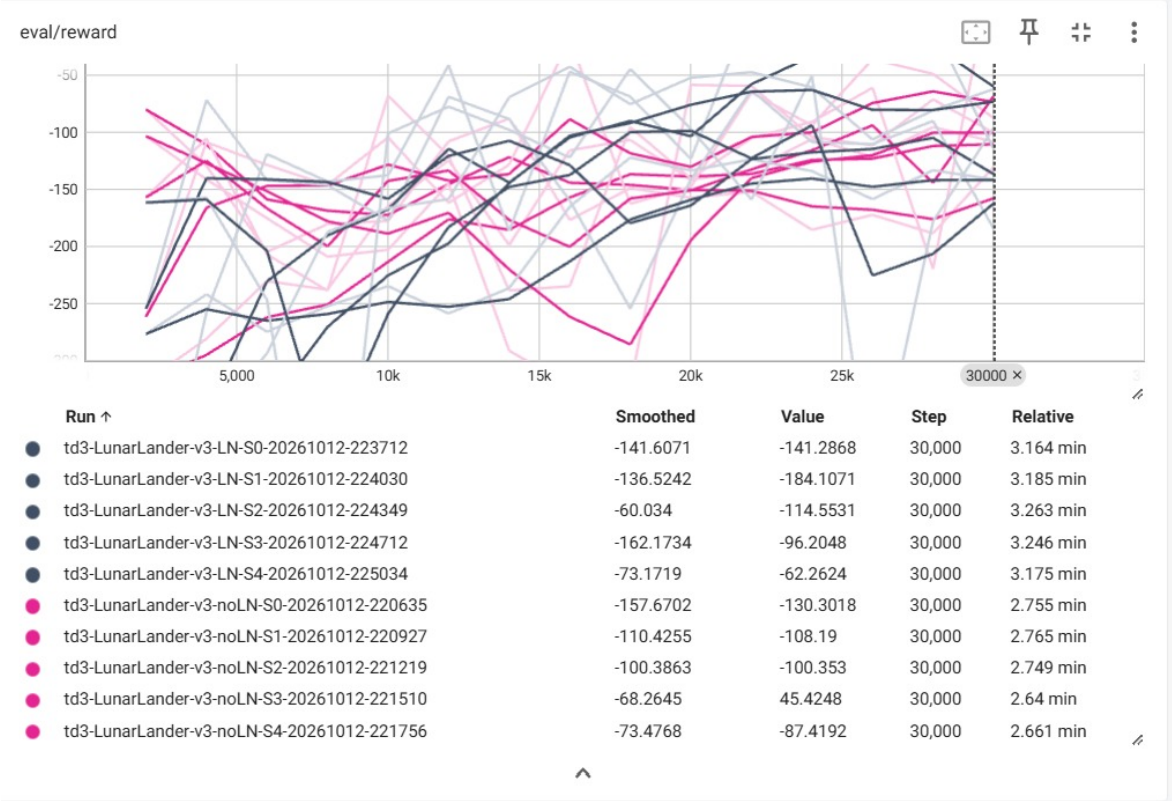
LN was therefore associated with lower final overestimation and critic loss, but not with better average final reward. With five seeds and one Monte Carlo rollout per checkpoint, these results remain descriptive; they do not establish a causal relationship between bias and performance.

# 3 TD3 with and without layer normalization

TD3 used the same shared hyperparameters as DDPG. The actor and target networks were updated every two critic updates policy_delay $= 2$. Target-policy noise had a standard deviation of 0.2 and was clipped to $[-0.5, 0.5]$. Table 3 summarizes all six recorded measures from the five seeds.

**Learning performance.** Just as for DDPG, reward curves fluctuate considerably (Figure 3). The final mean reward was $-76.2$ without LN and $-119.7$ with LN. Final rewards ranged from $-130.3$ to 45.4 without LN, compared with $-184.1$ to $-62.3$ with LN.

Table 3: Final checkpoint at 30,000 steps for TD3: mean ± sample standard deviation across five seeds. The loss entries describe the final training update.




| Run | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| td3-LunarLander-v3-LN-S0-20261012-223712 | -141.6071 | -141.2868 | 30,000 | 3.164 min |
| td3-LunarLander-v3-LN-S1-20261012-224030 | -136.5242 | -184.1071 | 30,000 | 3.185 min |
| td3-LunarLander-v3-LN-S2-20261012-224349 | -60.034 | -114.5531 | 30,000 | 3.263 min |
| td3-LunarLander-v3-LN-S3-20261012-224712 | -162.1734 | -96.2048 | 30,000 | 3.246 min |
| td3-LunarLander-v3-LN-S4-20261012-225034 | -73.1719 | -62.2624 | 30,000 | 3.175 min |
| td3-LunarLander-v3-noLN-S0-20261012-220635 | -157.6702 | -130.3018 | 30,000 | 2.755 min |
| td3-LunarLander-v3-noLN-S1-20261012-220927 | -110.4255 | -108.19 | 30,000 | 2.765 min |
| td3-LunarLander-v3-noLN-S2-20261012-221219 | -100.3863 | -100.353 | 30,000 | 2.749 min |
| td3-LunarLander-v3-noLN-S3-20261012-221510 | -68.2645 | 45.4248 | 30,000 | 2.64 min |
| td3-LunarLander-v3-noLN-S4-20261012-221756 | -73.4768 | -87.4192 | 30,000 | 2.661 min |

Figure 3: Evaluation reward versus environment steps of TD3. Bold and faint traces show TensorBoard-smoothed and raw values, respectively.

| Measure | Without LN | With LN |
|---|---|---|
| Evaluation reward | $-76.2 \pm 69.7$ | $-119.7 \pm 46.1$ |
| Predicted $Q(s_0, \pi(s_0))$ | $0.3 \pm 7.8$ | $0.5 \pm 7.8$ |
| Monte Carlo return $G_0$ | $11.0 \pm 13.9$ | $11.0 \pm 13.4$ |
| Estimated bias $Q - G_0$ | $-10.6 \pm 13.3$ | $-10.5 \pm 16.7$ |
| Actor loss | $23.6 \pm 11.8$ | $20.4 \pm 15.1$ |
| Critic loss | $38.6 \pm 61.3$ | $14.5 \pm 10.6$ |

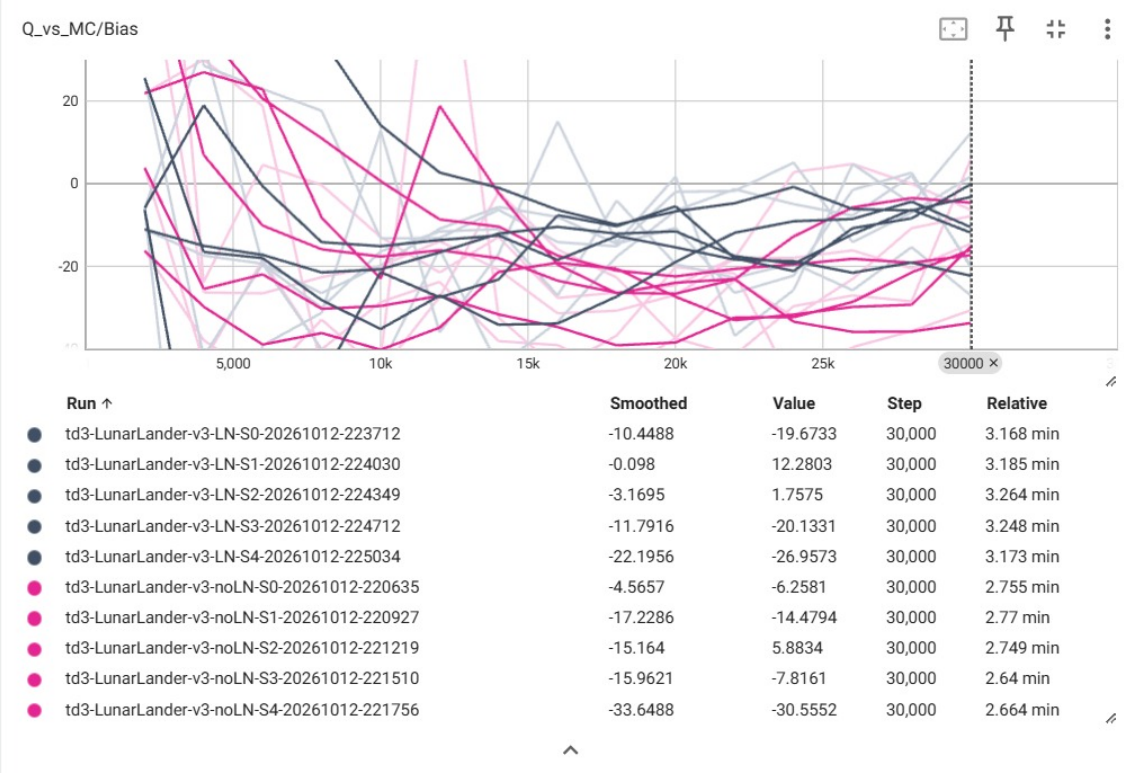
**Q-values and Monte Carlo returns.** Contrarily to DDPG, we obtain Q-values that are way closer to 0 with TD3. Predicted Q-values' final mean was 0.342 without LN and 0.47 with LN. Monte Carlo returns' final mean was 10.99 without LN and 11.018 with LN.

**Overestimation bias.** This time, the overestimation bias $Q - G_0$ alternated between positive and negative values. (Figure 4). Its mean was $-10.6$ without LN and $-10.5$ with LN, with similar mean values.

**Actor and critic losses.** The actor loss' final mean was 23.6 without LN and 20.4 with LN. The critic loss, which is actually the sum of the loss of each critic, showed frequent spikes. The mean loss of the critic is 38.6 without LN and 14.5 with $LN$.

# 4 Comparison and conclusion

At each evaluation, we estimated bias as $Q(s_0, \pi(s_0)) - G_0$ using one rollout without exploration noise, reset with seed $s + 1000$. Here, $s$ is the training seed and $G_0$ the discounted return from the same initial state and




| Run ↑ | Smoothed | Value | Step | Relative |
|---|---|---|---|---|
| td3-LunarLander-v3-LN-S0-20261012-223712 | -10.4488 | -19.6733 | 30,000 | 3.168 min |
| td3-LunarLander-v3-LN-S1-20261012-224030 | -0.098 | 12.2803 | 30,000 | 3.185 min |
| td3-LunarLander-v3-LN-S2-20261012-224349 | -3.1695 | 1.7575 | 30,000 | 3.264 min |
| td3-LunarLander-v3-LN-S3-20261012-224712 | -11.7916 | -20.1331 | 30,000 | 3.248 min |
| td3-LunarLander-v3-LN-S4-20261012-225034 | -22.1956 | -26.9573 | 30,000 | 3.173 min |
| td3-LunarLander-v3-noLN-S0-20261012-220635 | -4.5657 | -6.2581 | 30,000 | 2.755 min |
| td3-LunarLander-v3-noLN-S1-20261012-220927 | -17.2286 | -14.4794 | 30,000 | 2.77 min |
| td3-LunarLander-v3-noLN-S2-20261012-221219 | -15.164 | 5.8834 | 30,000 | 2.749 min |
| td3-LunarLander-v3-noLN-S3-20261012-221510 | -15.9621 | -7.8161 | 30,000 | 2.64 min |
| td3-LunarLander-v3-noLN-S4-20261012-221756 | -33.6488 | -30.5552 | 30,000 | 2.664 min |

Figure 4: Estimated bias $Q - G_0$ versus environment steps for the ten runs of TD3. Each raw observation uses one Monte Carlo episode. Some peaks exceed the supplied plot's vertical range.

| Algorithm | LN | Final reward | Bias $(Q - G_0)$ |
|---|---|---|---|
| DDPG | No | $-99.7 \pm 85.2$ | $55.6 \pm 31.1$ |
| DDPG | Yes | $-133.1 \pm 77.3$ | $26.3 \pm 13.9$ |
| TD3 | No | $-76.2 \pm 69.7$ | $-10.6 \pm 13.3$ |
| TD3 | Yes | $-119.7 \pm 46.1$ | $-10.5 \pm 16.7$ |

Table 4: Final results at 30,000 steps: mean ± standard deviation across five seeds.

action. TD3 used critic 1. TD3 obtained better average rewards in our runs. DDPG overestimated returns, while TD3 underestimated them on average. LN reduced DDPG's overestimation but did not improve the average reward of either algorithm. Results varied considerably across seeds, so these findings should not be generalized.

# AI use statement

The LLMs we used for this project are:

**Experimental design:**

- [ ] No LLM was used to design the experiments
- [ ] We questioned an LLM to help us design the experiments
- [x] An LLM designed X% of the experiments, with X =
- [ ] An LLM designed all the experiments on its own

**Coding:**

- [ ] No LLM was used to code the experimental study
- [x] We questioned an LLM to help us code the experimental study
- [ ] An LLM produced X% of the code, with X =
- [ ] An LLM produced all the code, we checked it
- [ ] An LLM produced all the code, we didn't check it

**Writing:**

- [ ] No LLM was used to write the report
- [ ] We wrote the report, an LLM polished it
- [x] An LLM participated in writing the report, up to X%, with X = 50
- [ ] An LLM wrote the report, we read and eventually corrected it
- [ ] An LLM wrote the report, we did not read it